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Multiple Qubits (Lecture 3)

Course: Survey of Quantum Technologies & Applications (PH 40001)

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Tensor Product

- When we have multiple qubits, we write their states as a tensor product $\otimes$.
- For example, two qubits, both in the $|0\rangle$ state, are written as $|0\rangle \otimes |0\rangle$.

$$|0\rangle \otimes |0\rangle \rightarrow |00\rangle$$

Same physical state, but compressed notation

Two-qubits in Z-basis

- With two qubits $|q_1 q_0\rangle$, the Z-basis is $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.
- A general state is a superposition of these basis states:

$$|\psi\rangle = c_0|00\rangle + c_1|01\rangle + c_2|10\rangle + c_3|11\rangle$$

- If we measure these two qubits in the Z-basis, we get,
  - $|00\rangle$ with probability $|c_0|^2$,
  - $|01\rangle$ with probability $|c_1|^2$,
  - $|10\rangle$ with probability $|c_2|^2$,
  - $|11\rangle$ with probability $|c_3|^2$
- Thus, $|c_0|^2 + |c_1|^2 + |c_2|^2 + |c_3|^2 = 1$

Three-qubits in Z-basis

- With three qubits $|q_2q_1q_0\rangle$, there are eight Z-basis states.

$\|000\rangle$	$\|001\rangle$	$\|010\rangle$	$\|011\rangle$
$\|100\rangle$	$\|101\rangle$	$\|110\rangle$	$\|111\rangle$

Kronecker Product

- In linear algebra, the tensor product is simply the Kronecker product.

$$|00\rangle = |0\rangle|0\rangle = |0\rangle \otimes |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

$$|01\rangle = |0\rangle|1\rangle = |0\rangle \otimes |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

Kronecker Product

- In linear algebra, the tensor product is simply the Kronecker product.

$$|10\rangle = |1\rangle|0\rangle = |1\rangle \otimes |0\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} .$$

$$|11\rangle = |1\rangle|1\rangle = |1\rangle \otimes |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} .$$

Kronecker Product

$$c_0|00\rangle + c_1|01\rangle + c_2|10\rangle + c_3|11\rangle = \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{pmatrix}.$$ For two qubits $|q_1 q_0\rangle$

$$\sum_{j=0}^{7} c_j|j\rangle = c_0|0\rangle + c_1|1\rangle + \cdots + c_7|7\rangle = \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_7 \end{pmatrix}.$$ For three qubits $|q_2 q_1 q_0\rangle$

$$\sum_{j=0}^{N-1} c_j|j\rangle = c_0|0\rangle + c_1|1\rangle + \cdots + c_{N-1}|N-1\rangle = \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_{N-1} \end{pmatrix}$$ For N qubits $|q_{N-1} \ldots q_1\ q_0\rangle$

Measuring Individual Qubits

Say we have two qubits in the state

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|01\rangle + \frac{\sqrt{3}}{4}|10\rangle + \frac{1}{4}|11\rangle.$$

If we measure both qubits, we would get $|00\rangle$ with probability $1/2$, $|01\rangle$ with probability $1/4$, $|10\rangle$ with probability $3/16$, or $|11\rangle$ with probability $1/16$.

Now, instead of measuring both qubits, let us only measure the left qubit. This yields $|0\rangle$ or $|1\rangle$ with some probabilities, and the state collapses to some state, so the outcomes are

$|0\rangle$ with some probability, and the state collapses to something,

$|1\rangle$ with some probability, and the state collapses to something.

Measuring Individual Qubits

The probability of getting $|0\rangle$ when measuring the left qubit is given by the sum of the norm-squares of the amplitudes of $|00\rangle$ and $|01\rangle$, since those both have the left qubit as $|0\rangle$. That is, the probability of getting $|0\rangle$ is

$$\left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{1}{2}\right|^2 = \frac{3}{4}.$$

Similarly, if the outcome is $|1\rangle$, then from the $|10\rangle$ and $|11\rangle$ states, the probability is

$$\left|\frac{\sqrt{3}}{4}\right|^2 + \left|\frac{1}{4}\right|^2 = \frac{1}{4}.$$

Then, the results of the measurement are:

$|0\rangle$ with probability $\frac{3}{4}$, and the state collapses to something?,

$|1\rangle$ with probability $\frac{1}{4}$, and the state collapses to something?.

Measuring Individual Qubits

Now for the states after measurement, if the outcome is $|0\rangle$, then the state collapses to the parts where the left qubit is $|0\rangle$, so it becomes

$$A\left(\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|01\rangle\right), \quad A = \frac{1}{\sqrt{|c_0|^2+|c_1|^2}}, c_0 = \frac{1}{\sqrt{2}} \quad c_1 = \frac{1}{2}$$

where A is a normalization constant. Similarly, if the outcome is $|1\rangle$, then the state collapses to the terms where the left qubit is $|1\rangle$, so it becomes

$$B\left(\frac{\sqrt{3}}{4}|10\rangle + \frac{1}{4}|11\rangle\right). \quad B = \frac{1}{\sqrt{|c_2|^2+|c_3|^2}}, c_2 = \frac{\sqrt{3}}{4} \quad c_3 = \frac{1}{4}$$

where B is a normalization constant.

Measuring Individual Qubits

Normalizing these, we get $A = 2/\sqrt{3}$ and $B = 2$, so measuring the left qubit yields

$$|0\rangle \text{ with probability } \frac{3}{4}, \text{ and the state collapses to } \sqrt{\frac{2}{3}}|00\rangle + \frac{1}{\sqrt{3}}|01\rangle,$$

$$|1\rangle \text{ with probability } \frac{1}{4}, \text{ and the state collapses to } \frac{\sqrt{3}}{2}|10\rangle + \frac{1}{2}|11\rangle.$$

We can apply these ideas to any number of qubits.

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